

Computing Performance Indicators in Detecting Shipping Collision

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ABSTRACT

The need to store and analyze movement data is continuously growing as a consequence of the huge availability of data being collected through GPS receivers, mobile devices, sensors, and others technologies. This paper presents an approach for the storage and analysis of maritime transportation data in a spatial data warehouse. The aim is to identify near-miss situations, which are the contexts where collisions between ships may occur. Moreover, a key performance indicator is proposed to compute the target percentage of safe situations between ships. The proposed spatial data warehouse was modeled, implemented and loaded with the data collected by The Netherlands Coastguard, and includes data of shipping movements collected by AIS (Automatic Identification System) base stations. Through the SOLAP analysis of this data set, and taking a sample safety distance of 50 meters between ships, it was possible to verify that the percentage of safe situations is of 92%, going beyond the defined target limit of 90%. The results are promising at the conceptual level and demonstrate the need for further development of key performance indicators for analyzing large movement data sets.

Categories and Subject Descriptors

H.2.8 [Information Systems]: Database Applications – Spatial databases and GIS.

General Terms

Algorithms, Management, Measurement.

Keywords

Movement data, spatial data warehouse, spatial on-line analytical processing, performance indicators.

1. INTRODUCTION

In order to achieve competitive advantage, organizations have always focused on Business Performance Management (BPM) for identifying the key performance indicators that contribute most to their success [1]. BPM indicators help organizations to become

more focused, aligned and effective [2]. Examples of key performance indicators include “Sales Values”, “Graduation Rates”, “Number of Infected”, among others, which can be associated with making, or not, profit. Over the last decade organizations started to collect and store huge amounts of movement data related to their day-by-day activities. However, analytical procedures have not been taken in order to analyze these movement data sets and to make available a broad range of key performance indicators that can guide an organization in pursuing its goals.

Spatial data warehouses have become a topic of growing interest in recent years due to the proliferation of movement data sets that are being collected from various sources such as GPS-receivers, communication media, online social networks and other geo-spatial applications [3]. The logical design of a spatial data warehouse is based on multidimensional data cube models that provide support for spatial data dimensions and spatial measures. Dimensions represent the analysis axes, while measures are the variables being analyzed against the different dimensions. The implementation of SOLAP (Spatial On-line Analytical Processing) tools can be achieved through solutions that are OLAP dominant, GIS (Geographical Information Systems) dominant, or both, in a mixed solution [4]. Those tools are powerful decision-making instruments as they allow users to explore movement data in a user friendly application and to formulate *ad-hoc* queries on these data.

When dealing with business performance indicators, SOLAP emerges as an enabling technology, providing the conceptual framework and the technological infrastructure for supporting the decision making process within an organization that requires measures that can serve as a vital decision-making tool. These measures provide information about movement behaviour that can be used to make improvements in domains like safety.

This paper addresses the maritime safety problem through the analysis of data collected by The Netherlands Coastguard. The data set includes data of shipping movements registered by AIS (Automatic Identification System) base stations. The maritime safety problem is becoming more and more important with the increasing number of pleasure boats and the growing density of sea traffic that increases the risk of collisions [5].

The contribution of this paper is two-fold: (a) define and compute a BPM performance indicator for analyzing movement data sets using SOLAP tools; (b) evaluate such an indicator in the study of near-miss situations where potential dangerous collisions may occur in maritime transportation.

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This paper is organized as follows. Section 2 introduces the concepts associated to business performance management and the key performance indicators that can be defined to measure organization's performance. Section 3 describes the main concepts associated to spatial data warehouses and spatial OLAP cubes. Section 4 presents the spatial data warehouse proposed in this work, which implements a cube that is analyzed in section 5. Section 6 concludes with some remarks and guidelines for future work.

2. BUSINESS PERFORMANCE MANAGEMENT

Business Performance Management (BPM) usually takes place at three different levels in organizations: strategic, tactical and operational [2,6]. At the strategic level, BPM makes available performance metrics to executives and senior managers. These metrics can be presented making use of scorecards or other tools [2]. At the tactical level, BPM indicators make use of analytical tools to analyze business trends, comparing actual metrics with past metrics or to identify business patterns. For this, OLAP and Data Mining tools have been extensively used in the past [7]. For the operational level, BPM indicators provide information about the organization day-by-day activities, supporting the accomplishment of the operational processes.

BPM is achieved through a continuous process of analysis of business measures, which allows the evaluation of business processes performance. Performance requirements on business processes are specified making use of Key Performance Indicators (KPI) that have target values which must be reached in a certain period [1]. KPIs reflect and derive from the organizational goals. Those goals guide the organization's efforts, support the distribution of resources and focus the organizations on success [8].

In our approach, each KPI was defined in such a way that makes possible its further analysis using OLAP or SOLAP tools. Our criteria were based on the commonly used reference that indicators must be SMART (*Specific, Measurable, Attainable, Realistic and Time-sensitive*) [8]. This implies: i) it has to be clear what the indicator describes, so it must be specific; ii) it has to be possible to measure a current value and be comparable to a target value; iii) it has to be possible to achieve the target value; iv) it has to be realistic in a particular working environment; and v) it has to be related to a specific time frame, which is the period in which it will be measured.

Finally, and as suggested by [1], our proposed lifecycle comprises the definition, measuring, analysis and report phases. During definition, indicators are identified and described, and target values are defined. After definition, indicators' values have to be measured. Analysis comprises the comparison of KPIs values with the defined measures.

3. SPATIAL DW AND SPATIAL OLAP

A Spatial Data Warehouse (SDW) can be defined as a large, subject-oriented, repository of data integrated from a variety of sources with an associated timeline. Since the pioneering work of Han et al. [9], several models have been proposed in the literature aiming at extending the classical data warehouse models with spatial concepts (turning into a SDW model) and the OLAP tools with spatial operators.

When adding spatiality to dimensions, most of the proposals follow or are based on the approach of Stefanovic et al. [10] and Rivest et al. [4] that distinguish three types of dimensions based on the spatial references of the members: i) descriptive dimensions without spatial attributes (non-geometric spatial dimension); ii) mixed spatial dimensions, containing both spatial and non-spatial attributes; and, iii) spatial dimensions containing only spatial attributes (geometric dimensions).

In the work presented in this paper two types of dimensions are used: mixed spatial dimensions, with spatial and non-spatial attributes, and descriptive dimensions in the case of the Time dimension.

Spatial attributes are associated with objects to which position on space is relevant. Depending on the type of spatial attribute we are dealing with, objects can geometrically be represented by points, lines or polygons. Those spatial attributes, when included in a SDW, allow the analysis of the fact measures looking at the geographical space where the events took place. Spatial and non-spatial attributes can integrate hierarchies (spatial or non-spatial) that can be used to do a more aggregated (*roll-up*) or more detailed data analysis (*drill-down*).

As it can be seen in the following section, our research work deals with point type objects that represent the position of moving objects over time. Several points are analyzed using spatial operators (e.g. intersection and distance) in order to verify their proximity in space and time. This allows the identification of measures like minimum distance between ships, minimum time difference between ships, number of ships inside a region, and others measures. After that, they are integrated in a fact table that is linked to two dimension tables: the Time dimension and the Spatial dimension. The spatial dimension includes attributes describing the several regions in which the analyzed space was split and to which the calculated measures are associated. These measures are later analyzed using traditional OLAP operations, like drill-down and slice, which allow a deeper understanding on data.

4. THE KPI LIFECYCLE IN A SPATIAL DATA WAREHOUSE

The data set presented in this section was collected by The Netherland Coastguard, and includes data of shipping movements collected by AIS base stations. The Maritime Research Institute (MARIN) in Netherlands receives the fused data for use in safety assessment studies with respect to shipping. MARIN has anonymized a week of data, which has been used in this research. This data set contains raw data with a time interval of 60 seconds between readings, which makes it suitable for the analysis of movement behaviors, like drifting ship or near-miss situations. In this paper the research assumption is that the MARIN organization is interested in measuring a performance indicator that can provide information of the high/low occurrence of near-miss situations, i.e. situations where ships have a very near collision.

For each ship, information like *Ship ID*, *Ship Type*, *Ship Size*, *Latitude*, *Longitude*, *Heading Rate of Turn*, *Speed Over Ground* and *Parse time* is available, as depicted in Figure 1.

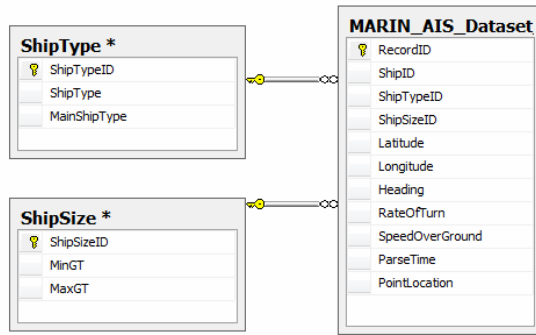


Figure 1. A sub-set of the available attributes

In one week of data, from 1 to 8 January 2009, 315 794 different records were collected (some duplicate readings available in the original data source were removed). The spatial distribution of the readings is shown in Figure 2.

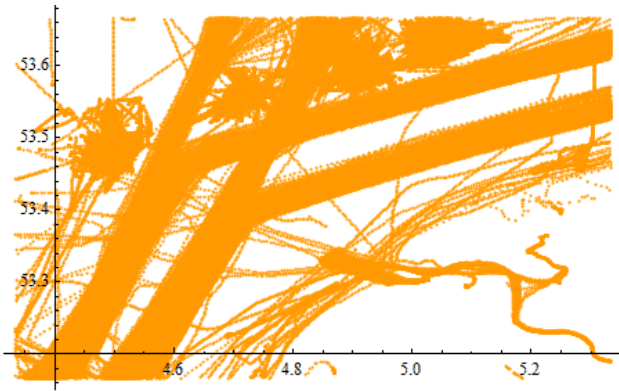


Figure 2. Spatial distribution of the ships

Looking at the available data and the objective of the BPM analysis (i.e. identify situations where a collision between ships can occur), the definition of the SDW schema was carried out. The geographical space shown in Figure 2 was sliced into several cells. In this case, 50 slices were considered for Latitude and another 50 slices for Longitude, which originated a grid with 2500 cells. After the division of the space, it was verified the number of different ships inside each one of these cells, and the minimum distance and maximum distance between two ships in the cell. Also, the time differences (in seconds) between ships inside a cell were calculated. From these values, the minimum and maximum time differences were considered. To complement this, the information about the minimum and maximum speed over ground of the ships inside a cell was also identified. Through this process it is possible to verify if at a given time interval a cell is crowded of ships and also if those ships are near each other in space (distance) and time (Figure 3). The distances between ships are calculated considering the positions given by the AIS readings.

If a cell, representing a specific region in the analyzed area, is crowded and the ships are near in space and time then we could be in a potential collision situation. After identifying those situations and taking the appropriate actions, MARIN researchers can verify, in future analyzes, if the number of near-miss situations decreased and, if yes, if the reached values are in accordance with the defined performance indicators.

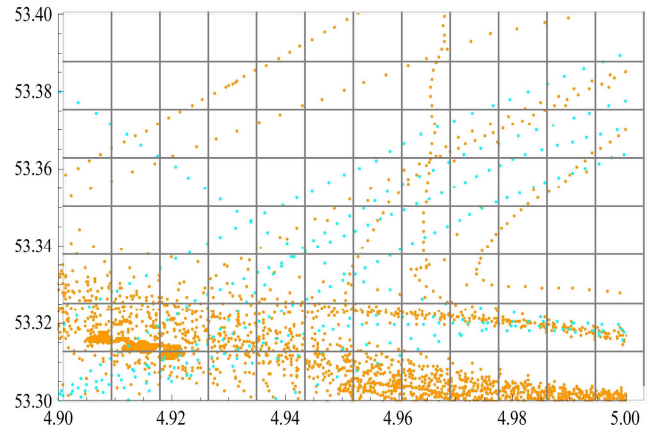


Figure 3. Split of the geographical space in cells

All the raw data was stored in a relational database in SQL Server 2008© (the relational schema was already presented in Figure 1). All the processing needed to collect the measures that will be included in the SDW was done using SQL Scripts written in Transact-SQL (next section will detail this component).

The proposed multidimensional data cube model includes two dimension tables and one fact table. The dimensional tables are *Spatial Cells*, which includes all the detail about the created cells, and *Time*, to be possible the analysis of data through time. The fact table is named *Near-Miss Situations* and includes the foreign keys to the dimensions and the measures identified as relevant to the analysis of potential dangerous situations. Figure 4 presents the multidimensional data cube model proposed for the SDW. This model is shown following the ADAPT™ (*Application Design for Analytical Processing Technologies*) notation [11]. The fact table *Near-Miss Situations* presents two different links to the time dimension as all aggregates were calculated in a time interval of 5 minutes. This means that for each cell we have the start time represented by the minimum time reading of a ship position in that cell and the end time as the maximum time reading of a ship in the same cell, considering a maximum interval of 300 seconds between these two readings.

After the presentation of the SDW model, we need to define the performance indicator that will be considered to measure if the actions performed to decrease the number of potential dangerous situations in this application domain were successful or not. Table 1 presents the defined key performance indicator and the target value to be achieved.

Table 1. Definition of a Key Performance Indicator

Key Performance Indicator	Value
% of Safe Situations	$\geq 90\%$

(the percentage of safe situations will be calculated considering a minimum safety distance of 50 meters between ships)

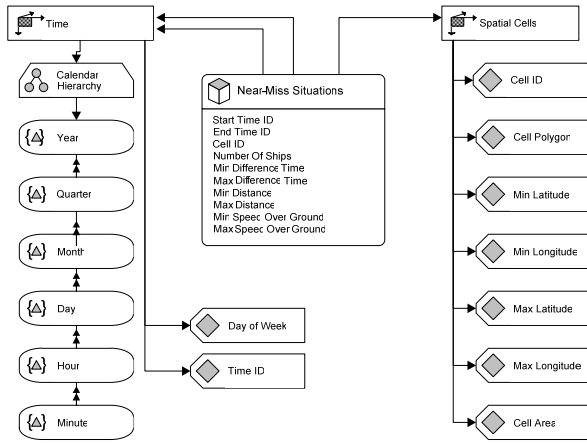


Figure 4. SDW Model

5. NEAR-MISS SITUATIONS CUBE

After the definition and implementation of the multidimensional data cube model in the SQL Server 2008 platform, a set of scripts were written in Transact-SQL (T-SQL) to implement the ETL (*Extraction, Transformation and Loading*) process. The first step was associated with the calculation of the several indicators present in a cell. For each cell, having a maximum time interval of 5 minutes (300 seconds), the following indicators were calculated: i) the number of ships; ii) the minimum and maximum time difference between two ships; iii) the minimum and maximum distance between two ships; and, iv) the minimum and maximum speed over ground.

Through this process, the implemented T-SQL script (Figure 5) allowed the creation of an auxiliary table with 2500 rows. This table integrates the several measures under analysis.

```

/* Calculating and storing the pre-processed data */
Use MarinData
DECLARE @i int;
DECLARE @j int;
DECLARE @dTime int;
DECLARE @nShips int;
DECLARE @p1 geography;
DECLARE @maxTime int;
DECLARE @minTime int;

SET @minTime=(Select MIN(CAST(ParseTime As int)) from MARIN_AIS_Dataset_2a)
SET @maxTime=(Select MAX(CAST(ParseTime As int)) from MARIN_AIS_Dataset_2a)

SET @dTime=300 /* Five Minutes */
SET @i=1;
SET @j=(SELECT COUNT(*) From Cell)
WHILE @i<=@j
BEGIN
    SET @p1 = (Select CellPolygon
                From Cell
                Where CellID=@i)

    Insert Into auxData
    Select Count(Distinct(pl.ShipID)) As numShips,
    Min((ABS(CAST(pl.ParseTime As int)-CAST(p2.ParseTime As int)))) As MinTimeDiff,
    Max((ABS(CAST(pl.ParseTime As int)-CAST(p2.ParseTime As int)))) As MaxTimeDiff,
    Min(pl.PointLocation.STDistance(p2.PointLocation)) As MinDist,
    Max(pl.PointLocation.STDistance(p2.PointLocation)) As MaxDist,
    Min(CAST(pl.SpeedOverGround As float)) As MinSpeed,
    Max(CAST(pl.SpeedOverGround As float)) As MaxSpeed,
    DatePart(yy,Dateadd(ss,MIN(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tYear,
    DatePart(mm,Dateadd(ss,MIN(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMonth,
    DatePart(dd,Dateadd(ss,MIN(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMinDay,
    DatePart(hh,Dateadd(ss,MIN(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMinHour,
    DatePart(mi,Dateadd(ss,MIN(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMinMinute,
    DatePart(dd,Dateadd(ss,MAX(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMaxDay,
    DatePart(hh,Dateadd(ss,MAX(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMaxHour,
    DatePart(mi,Dateadd(ss,MAX(CAST(pl.ParseTime As int)-@minTime), '2009/01/01')) As tMaxMinute
    From MARIN_AIS_Dataset_2a p1, MARIN_AIS_Dataset_2a p2
    Where ABS(CAST(pl.ParseTime As int)-CAST(p2.ParseTime As int))<=@dTime
    And p1.ShipID<>p2.ShipID
    And p1.RecordID IN (Select RecordID
                        From MARIN_AIS_Dataset_2a
                        Where @p1.STIntersects(PointLocation)=1)
    And p2.RecordID IN (Select RecordID
                        From MARIN_AIS_Dataset_2a
                        Where @p1.STIntersects(PointLocation)=1)

    SET @i=@i+1
END
GO

```

Figure 5. Script to identify the measures

The process of creating and loading the SDW was achieved using another T-SQL script (Figure 6) that creates the SDW tables and loads all the data available in the auxiliary table and in the related tables (relational data model). This script creates the several tables present in the multidimensional model (*Time, Spatial Cells and*

Near-Miss Situations) and does the proper processing to fill each one of the dimensions (*Time and Spatial Cells*) and the fact table (*Near-Miss Situations*).

```

-- Source Databases
USE MarinData
USE MarinDW

-- dimSpatialCells
-- Creating the dimension with the Cell's information
CREATE TABLE [MarinDW].[dbo].[dimSpatialCells]
(
    CellID int Primary Key,
    CellPolygon geography,
    MinLat float,
    MinLog float,
    MaxLat float,
    MaxLog float,
    CellArea float)
GO

-- Loading data to the dimension
INSERT INTO [MarinDW].[dbo].[dimSpatialCells]
SELECT *
FROM [MarinData].[dbo].[Cell]
GO

-- dimTime
-- Creating Time Dimension
CREATE TABLE [MarinDW].[dbo].[dimTime]
(
    TimeID int IDENTITY(1,1) PRIMARY KEY,
    TDate date,
    TMinute int,
    THour int,
    TDay int,
    TDayOfWeek varchar(10),
    TMonth varchar(10),
    TQuarter int,
    TYear char(4),
)
GO

-- Loading data into the Time Dimension
DECLARE @OrderDate DATETIME
DECLARE @OrderEnd DATETIME
DECLARE @HourStart INT
DECLARE @HourEnd INT
DECLARE @MinStart INT
DECLARE @MinEnd INT
SET @OrderDate = '2009/01/01'
SET @OrderEnd = '2009/01/08'
SET @HourStart=0
SET @HourEnd=23
SET @MinStart=0
SET @MinEnd=59
WHILE @OrderDate <= @OrderEnd
BEGIN
    SET @HourStart=0
    WHILE @HourStart<=@HourEnd
    BEGIN
        SET @MinStart=0
        WHILE @MinStart<=@MinEnd
        BEGIN
            INSERT INTO [MarinDW].[dbo].[dimTime]
            ([TDate],[TMinute],[THour],[TDay],[TDayOfWeek],
            [TMonth],[TQuarter],[TYear])
            VALUES (@OrderDate, DATEPART(mi,DATEADD(mi,@MinStart,@OrderDate)),
            DATEPART(hh,DATEADD(hh,@HourStart,@OrderDate)),DATEPART(dd,@Orderdate),
            DATENAME(dw,@Orderdate), DATENAME(mm,@Orderdate),
            DATEPART(qq,@Orderdate), DATEPART(yyyy,@Orderdate))
            SET @MinStart=@MinStart+1
        END
        SET @HourStart=@HourStart+1
    END
    SET @OrderDate = @OrderDate + 1
END

-- Fact Table: NearMissSituations
-- Creating the fact table
CREATE TABLE [MarinDW].[dbo].[NMSituations] (
    StartMinuteID int,
    EndMinuteID int,
    CellID int,
    NumberOfShips int,
    MinDiffTime int,
    MaxDiffTime int,
    MinDistance float,
    MaxDistance float,
    MinSpeedOverGround float,
    MaxSpeedOverGround float,
    PRIMARY KEY (StartMinuteID, EndMinuteID, CellID),
)
GO

-- Loading the fact table
INSERT INTO [MarinDW].[dbo].[NMSituations]
([StartMinuteID],[EndMinuteID],[CellID],[NumberOfShips],[MinDiffTime],
[MaxDiffTime],[MinDistance],[MaxDistance],[MinSpeedOverGround],
[MaxSpeedOverGround])
SELECT dimTime1.TimeID, dimTime2.TimeID, auxData.CellID, auxData.numShips,
auxData.minDiffTime, auxData.maxDiffTime,
auxData.minDist, auxData.maxDist,
auxData.minSpeed, auxData.maxSpeed
FROM [MarinDW].[dbo].[dimTime] As dimTime1,
[MarinDW].[dbo].[dimTime] As dimTime2,
[MarinData].[dbo].[auxData]
WHERE auxData.numShips IS NOT NULL And
auxData.tMinDay=dimTime1.TDay And auxData.tMinHour=dimTime1.THour And
auxData.tMinMinute=dimTime1.TMinute And auxData.tMaxDay=dimTime2.TDay And
auxData.tMaxHour=dimTime2.THour And auxData.tMaxMinute=dimTime2.TMinute
GO

```

Figure 6. Creating and loading the SDW

After the loading process, the corresponding cube was created in the *SQL Server Business Intelligence Development Studio*. The cube model is shown in Figure 7.

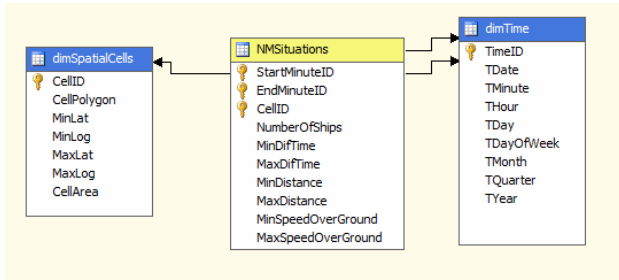


Figure 7. The obtained cube

All the available movement data can now be manipulated using traditional cube operations, like drill-down, roll-up, slice and dice, and pivot. The defined time hierarchy allows the analysis of the movement data using a minimum granularity of a minute and a maximum of one year (although at this moment only data of one week was loaded into the SDW). The defined hierarchy is as follows: Minute → Hour → Day → Month → Quarter → Year. Figure 8 presents a view of the available attributes and the cube area for data exploration.

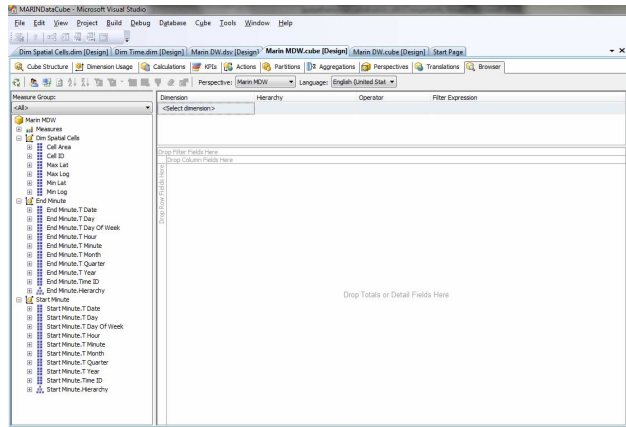


Figure 8. The available data and the cube browser

Starting by obtaining an overall picture of the available aggregated data, Figure 9 shows that for each day under analysis, and for each one of the cells that were used to split the geographical space, there is always a ship that is located in the same cell at the same time instant (*Min Dif Time*). The only exception is for day 2009-01-07 where a minimum time difference of 3 seconds is verified between two ships. It is worth to mention that for the several days, short distances between ships are verified. The minimum distance verified between two ships is of 6,6 meters.

T Date	Min Dif Time	Max Dif Time	Min Distance	Max Distance	Number Of Ships	NM Situations Count
2009-01-01	0	300	6.65732511472917	1626.36844680996	2466	152
2009-01-02	0	300	12.9598464948094	1630.99586597402	1617	98
2009-01-03	0	300	17.3291369444894	1568.51353870483	1406	146
2009-01-04	0	300	82.8724761923793	1564.480597325	193	43
2009-01-05	0	300	42.6093931154022	1518.98330050508	89	24
2009-01-06	0	300	25.9113479514725	1408.6908209803	219	69
2009-01-07	3	300	45.5351937074292	1534.70713735195	46	21
2009-01-08	0	300	6.63919962604116	1469.96475240947	56	27
Grand Total	0	300	6.63919962604116	1630.99586597402	6092	580

Figure 9. Summary of the obtained measures

If we look at the number of ships that are positioned in the same cell at a time interval of 5 minutes (300 seconds), we can see (Figure 9) that along the week 6092 ships were processed and aggregated in this data cube. The total number of cells with data in this time frame is of 580 (from a total of 2500). The maximum distance between two ships in 300 seconds varies from 1409 to 1631 meters.

Going down to get a deeper understanding on data, we verified that different days have different behaviors. If we look at day 2009-01-06 (Figure 10) we can see that the minimum speed over ground is always different from 0 denoting that none of these ships is stopped.

Dimension	Hierarchy		Operator	Filter Expression		
Start Minute		Start Minute.T Day	Equal	{6}		
<Select dimension>						
Drop Filter Fields Here						
Drop Column Fields Here						
T Hour	Min Dif Time	Max Dif Time	Min Distance	Max Distance	Number Of Ships	Min Speed Over Ground
0	0	300	228.707905729288	1163.54383833898	20	6.2
1	0	300	189.321190667922	1325.7229356805	29	1.9
10	0	300	236.06336386965	1379.72009739831	14	7.8
11	0	300	381.197429140294	1237.30976305601	12	8.2
12	243	300	592.249339801504	829.69202351143	4	10.9
14	120	300	533.986632448177	1213.69247130011	8	3.9
15	0	300	350.368507269524	1408.6908209803	10	8.9
16	0	297	126.86230739776	1141.47524595403	4	8.4
18	0	63	637.167492699434	1270.42539688654	10	20.2
19	3	300	499.290927468361	1016.37927677018	15	15
2	0	300	25.9113479514725	1240.36408519937	30	5.8
20	240	300	520.149198874074	919.104617338577	2	15.2
3	0	300	353.747944984488	1232.83878294845	18	3.7
4	0	174	714.792382856985	1294.27666149531	2	10.3
5	0	300	218.831694270375	1242.50303183953	18	8
7	0	300	168.936941339581	1264.4116254828	12	8.7
8	3	300	753.826686185442	1291.35729466098	8	9.1
9	54	300	502.693755378449	1159.88508896948	10	10.5
Grand Total	0	300	25.9113479514725	1408.6908209803	219	1.9

Figure 10. Cube for day 2009-01-06

However, day 2009-01-08 (Figure 11) has several hours with a minimum speed over ground of 0 or a value close to 0, maybe because several fishing ships were present during this day in this geographical space.

Dimension	Hierarchy	Operator	Filter Expression			
Start Minute	 Start Minute.T.Day	Equal	{8}			
<Select dimension>						
Drop Filter Fields Here						
Drop Column Fields Here						
T Hour	Min Dif Time	Max Dif Time	Min Distance	Max Distance	Number Of Ships	Min Speed Over Ground
10	0	186	841.423195055355	1469.96475240947	2	4.3
15	246	300	520.150455449845	741.953345171709	2	12.6
16	0	297	126.32963930301	1012.4992313763	4	2.1
18	0	300	22.7779861909139	913.23504233263	8	5.8
19	0	300	33.1846014400876	961.147865130488	8	0
20	0	300	6.63919962604116	934.170095986431	6	0.2
21	0	300	6.64189992158699	878.406758451351	6	0
22	0	300	6.64269584571825	653.362099326329	6	0
23	0	300	17.3281846971658	587.932008071827	4	0.2
5	0	300	214.85843864027	960.052180554648	4	0
8	0	300	22.8214001946199	934.280317604888	4	2.1
9	0	297	408.576651899203	993.127474587136	2	7.8
Grand Total	0	300	6.63919962604116	1469.96475240947	56	0

Figure 11. Cube for day 2009-01-08

One interesting cube is obtained for day 2009-01-07, where the time difference between ships is always different from 0 (Figure 12). In this day, few near-miss situations (where ships have a very near encounter) seem to be identified looking at the distance between ships. However, a possible near-collision can only be detected if the ships have routes that will intersect each other in a near future. These situations are out of the scope of this paper and will be addressed as future work.

Dimension	Hierarchy	Operator	Filter Expression			
Start Minute	 Start Minute.T Day	Equal	{7}			
<Select dimension>						
Drop Filter Fields Here						
Drop Column Fields Here						
T Hour	Min Dif Time	Max Dif Time	Min Distance	Max Distance	Number Of Ships	Min Speed Over Ground
1	117	297	593.568435379675	1117.6126804274	6	5.2
10	240	240	1308.65300473248	1308.65300473248	2	10.9
11	3	300	119.527411407467	1182.86821489671	10	9.9
16	117	180	943.028357168545	1088.58461058767	2	11.5
19	6	300	1105.15141667266	1105.15141667266	2	0
22	297	297	1065.43186488571	1065.43186488571	2	13.2
23	3	300	248.140873717539	1534.70713735195	14	0.2
3	3	300	45.5351937074292	985.76299465584	6	2.1
9	69	159	331.181167799624	746.839023067078	2	8.9
Grand Total	3	300	45.5351937074292	1534.70713735195	46	0

Figure 12. Cube for day 2009-01-07

Focusing now on the performance indicator that was defined to this maritime transportation domain, it will be verified if the

percentage of situations where the distance considered safe between ships is equal or higher than 90%. In this task we consider, as a sample value, that this safety distance must be equal or higher than 50 meters. Figure 13 shows the MDX (Multidimensional Expression Language) code implemented to calculate the key performance indicator taking into account the number of safe situations (*Safe* measure) and the number of situations under analysis (*NM Situations Count* measure).

```
/* Calculating the new measures */
CALCULATE;
CREATE MEMBER CURRENTCUBE.[Measures].[Safe]
AS IIF([Measures].[Min Distance]>=50,1,0),
NON_EMPTY_BEHAVIOR = ([NM Situations Count] ),
VISIBLE = 1, ASSOCIATED_MEASURE_GROUP = 'NM Situations';
CREATE MEMBER CURRENTCUBE.[Measures].[% Safe Situations]
AS [Measures].[Safe]/[NM Situations Count],
FORMAT_STRING = "Percent",
VISIBLE = 1;
```

Figure 13. MDX code for the calculation of the new measures

After the calculation of the percentage of safe situations, Figure 14 shows that this percentage is always higher than 90% except for day 2009-01-08, where the number of ships with a low distance between them is high.

T Day	T Hour	Min Distance	Number Of Ships	Safe	NM Situations Count	% Safe Situations
1		6.65732511472917	2466	142	152	0.934210526315789
2		12.9598464948094	1617	92	98	0.938775510204082
3		17.3291369444894	1406	137	146	0.938356164383562
4		82.8724761923793	193	43	43	1
5		42.6093931154022	89	23	24	0.958333333333333
6		25.9113479514725	219	68	69	0.985507246376812
7		45.5351937074292	46	20	21	0.952380952380952
8		6.63919962604116	56	9	27	0.333333333333333
Grand Total *		6.63919962604116	6092	534	580	0.920689655172414

Figure 14. Percentage of safe situations

Figure 15 shows a drill-down operation over this day. During night, mainly, a relevant number of ships in what could be considered a non-safe situation is verified. However, some ships are stopped and this may represent a normal situation. Further analyses and additional measures are needed to verify these cases in more detail.

T Day	T Hour	Min Distance	Number Of Ships	Safe	NM Situations Count	% Safe Situations
1		6.65732511472917	2466	142	152	0.934210526315789
2		12.9598464948094	1617	92	98	0.938775510204082
3		17.3291369444894	1406	137	146	0.938356164383562
4		82.8724761923793	193	43	43	1
5		42.6093931154022	89	23	24	0.958333333333333
6		25.9113479514725	219	68	69	0.985507246376812
7		45.5351937074292	46	20	21	0.952380952380952
8	10	841.423195065355	2	1	1	1
	15	520.150455449845	2	1	1	1
	16	126.32963930301	4	2	2	1
	18	22.7779861909139	8	1	4	0.25
	19	33.1846014400876	8	0	4	0
	20	6.63919962604116	6	0	3	0
	21	6.64189992158699	6	0	3	0
	22	6.64269584571825	6	1	3	0.333333333333333
	23	17.3281846971658	4	0	2	0
	5	214.85843864027	4	2	2	1
	8	22.8214001946199	4	0	1	0
	9	408.576651899203	2	1	1	1
	Total *	6.63919962604116	56	9	27	0.333333333333333
Grand Total *		6.63919962604116	6092	534	580	0.920689655172414

Figure 15. Drill-down for day 2009-01-08

After the presentation of the obtained results, and looking at the exercise of defining a SDW, identifying the performance indicators and verifying the obtained values, it can be said that the procedures followed to ensure the safety of the ships were effective. In this example, an overall value of 92% was obtained. Nevertheless, it needs to be mentioned that the defined values and the assumptions that were made need to be considered as abstract values that were set to show how the concept of key performance indicator can be applied to applications that manipulate movement data and where business performance management concepts need to be considered in order to support the decision-making process on those application domains.

6. CONCLUSION

This paper presented the design and implementation of a spatial data warehouse able to store data of shipping movements. The collected raw data was extracted, transformed and loaded to the implemented data warehouse, making possible data analysis using OLAP operations. The defined measures were explored allowing the identification of near-miss situations, which are the contexts where collisions between ships may occur. A key performance indicator was also defined, computed and evaluated. The results obtained were promising in the application of business performance management concepts to the spatial domain and, particularly, to movement data sets.

As future work, new measures will be included in the spatial data warehouse. For example, a flag to identify if two ships are in a collision route is needed. Concerning the division made of the space, and the aggregates that were calculated for each cell, buffer zones around each cell need to be defined in order to verify if a ship, intersecting another cell, is too close of the ships present in the cell under analysis.

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